

Machine learning enhanced next-generation optical access networks—challenges and emerging solutions [Invited Tutorial]

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Optical access networks are envisioned to become increasingly complex as they support more and more diverse and immersive services, each with a different capacity, latency, and reliability need. While machine learning has been touted as a silver bullet that will intelligently manage network operations and resources to meet these demands, as it had been anticipated for core and metro networks, there exist various challenges that need to be addressed to progress machine learning models from research to production. In this tutorial, we first aim to motivate the continued push to advance optical access networks and rationalize the use of machine learning in these networks. We then highlight the challenges that are especially amplified due to the traffic dynamicity and heterogeneity, data scarcity, and computation-resource constraints of optical access networks. We discuss emerging machine learning approaches that are being explored to address these challenges. Finally, we consider a fast and self-adaptive machine learning enhanced dynamic bandwidth allocation scheme in an illustrative future use case of supporting immersive human-to-machine communications over the mobile fronthaul of next-generation mobile networks. © 2023 Optica Publishing Group

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1. INTRODUCTION

Over the past 30 years, optical access networks have continued to evolve to satisfy the ever-increasing bandwidth demand, customer numbers, and quality-of-service (QoS) requirements of fixed business and residential as well as 4G and 5G mobile network deployments [1,2]. In that regard, we have witnessed the growth of passive optical network (PON) technology in particular, with standards evolving from the early ITU-T G.984 Gigabit PON in 2003 [3] and IEEE 802.3ah Gigabit Ethernet PON (1G-EPON) in 2004 [4], through to the most recent ITU-T G.9804 High Speed PON [5] and IEEE 802.3ca 25G/50G-EPON [6]. Already, ongoing research in exploring coherent transmission, orthogonal frequency division multiplexing, and wavelength bonding, along with standardization activities, are driving PON capacities beyond 100 Gbps. According to [7], the global optical access market will continue to grow with a compound annual growth rate of 9.3% to reach an estimated USD22.4B by 2026 from USD12.9B in 2020.

Additionally, due to its cost, reach, and capacity benefits, PON technology is being considered a candidate optical transport solution of next-generation mobile networks as they evolve towards 6G [8,9]. At the same time, more and more innovative applications are starting to appear as fibers are

being pushed deeper and deeper towards the edge to directly connect with users and their devices [10]. In the near term, optical access networks are envisioned to support the delivery of a diverse and immersive set of services including mixed reality (XR), holographic communication, human-to-machine (H2M)/robot communications, Tactile Internet, and digital sensing. However, in supporting these diverse and immersive services, traffic on the network will become increasingly heterogeneous and dynamic across a range of different time scales. Moreover, each service will command a different capacity, latency, and reliability need. The confluence of the above factors has placed significant demands on the optical access segment to meet a range of QoS and quality-of-experience requirements [11].

Machine learning (ML) has become ubiquitous in many fields, and its exploration for application in optical communication and network fields to realize many system and network functionalities is no exception. To date, there exists a vast library of tutorial and survey papers that collectively provide a comprehensive overview of physical and/or networking layer ML applications, a small subset of which is listed in [12–19]. A majority of the tutorials and survey papers in the existing

literature are focused on applications in core and metro networks, with only a few concentrating on short-reach, e.g., [18] and access networks, e.g., [19].

Even fewer have investigated the justification of ML in superseding conventional approaches based on acquiring network or system domain knowledge. In fact, framing a problem to be solved using ML instead of conventional approaches and the selection of which ML technique to use depend on many factors [20–22]. These factors include deficiency in conventional approaches; clarity of the problem definition, feature definition, and metric or evaluation criteria; the sufficiency of representative data; and the existence of infrastructure to collect said data. As highlighted in [21,22], the argument to utilize ML holds (a) when there lacks a physics-based mathematical model due to insufficient domain knowledge (model deficit) or (b) when the algorithms to run known physics-based mathematical models are computationally complex and/or time costly (algorithm deficit). Even when a problem is deemed to be better solved through ML rather than the conventional approach, key open challenges in confidently progressing ML models from pure research to production still remain. Equally pertinent to highlight is that these challenges are amplified in optical access network environments due to their being highly cost sensitive, having limited computational resources, being a highly dynamic network environment, and lacking installed data acquisition/monitoring infrastructure and hence representative data. Moreover, the need to support real-time applications places further demands on ML models to adapt with high responsiveness.

In this tutorial, we aim to discuss these challenges, in particular those that are amplified by the traffic dynamicity and heterogeneity, data scarcity, and computation-resource constraints of optical access networks. Consequently, we will bring to attention emerging ML approaches that are currently being investigated to address these challenges. Further, acknowledging that ML, while promising, incurs learning times that constrain rapid response to network changes, hence affecting overall network latency performance, we propose a fast and self-adaptive (FSA) ML-enhanced dynamic bandwidth allocation (DBA) scheme that combines reinforcement and transfer learning to achieve self-adaptive and optimized bandwidth decisions with reduced learning time. Specifically, we consider an illustrative future use case of deploying a 10 Gbps PON to support immersive H2M traffic over the mobile fronthaul (MFH) of next-generation mobile networks.

The remainder of this tutorial is structured as follows. In Section 2, we provide a discussion on major consortia initiatives and industry trends that are driving PON technology advancements and motivating its adoption of ML. In Section 3, we present the factors contributing to the current revival of ML research, and some common ML terminology and categories of algorithms. In Section 4, we provide a summary of the open key challenges impeding the progress of ML research into production. In Section 5, we discuss emerging ML approaches and solutions that are currently being explored to address said challenges. In Section 6, we present our proposal of the rapid and self-adaptive DBA scheme, along with results from our performance evaluation. Section 7 summarizes this tutorial.

2. DRIVERS AND FUTURE TRENDS

In this section, the continuing need to advance optical access networks and to enhance their capabilities through machine intelligence is motivated through brief overviews of major consortium and industry trends and initiatives.

Fifth Generation Fixed Network. In 2020, the European Telecommunications Standards Institute (ETSI) launched its Industry Specification Group (ISG) Fifth Generation Fixed Network (F5G) initiative to extend the benefits of fixed optical networks to seed new applications in the business and residential market segments [10]. Core to F5G is the vision of “Fiber To Everywhere/Everything.” In that regard, the expectation is to rely on the complementary use of WiFi 6 to access the last few meters of the network. Specific use cases range from PON on-premise, passive optical local area network (LAN), multiple access aggregations over PON, 5G front-/mid-/backhaul optical transport, cloud virtual reality, enhanced traffic monitoring and network control, and PON for industrial manufacturing [23,24]. F5G is embodied by three network dimensions, namely, (a) enhanced fixed broadband (eFBB), which supports applications that require large bandwidth; (b) guaranteed reliable experience (GRE), which supports applications that require low latency; and (c) full fiber connection (FFC), which supports applications that require massive fiber connections. The reliance of one or more of these network dimensions will depend on the specific use case. The F5G network architecture comprises an underlay plane; a service plane; and a management, control, and analytics (MCA) plane, as illustrated in Fig. 1 [25]. The MCA plane is the intelligence center of F5G, relying on an artificial intelligence (AI) engine for data analytics and network diagnosis to autonomously coordinate the configuration and programmability of the underlay and services planes. Network resources are intelligently planned, coordinated, and mobilized to realize the three network dimensions mentioned above. In turn, the underlay plane consists of (a) customer premises networks, e.g., passive optical LAN, WiFi 6, Fiber-to-the-Room (FTTR); (b) optical access networks based on 10-gigabit-capable symmetric PON (XGS-PON) technology [26]; (c) an optical aggregation network, e.g., parallel Internet Protocol/Ethernet and optical transport network (OTN) fabrics; and (d) the optical core network. In terms of the service plane, it is decoupled from the underlay plane, providing service flexibility and supporting services with different service level agreements.

All-Photonics Network. The All-Photonics Network (APN) was proposed by NTT to provide ultra-realistic services comprising real-time ultra-high-definition video information and diverse, multi-sense content [27]. The underlying network of the APN is an optical full-mesh network that provides end-to-end optical paths for each service [28].

In the APN, the optical access network is directly connected to the optical backbone network through a photonic gateway (Ph-GW) that eliminates latency-contributing electronic processing delays, as shown in Fig. 2. A key enabling function of the Ph-GW is to remotely manage/control optical wavelength channels in the access network segment to facilitate end-to-end optical connections. The communication between

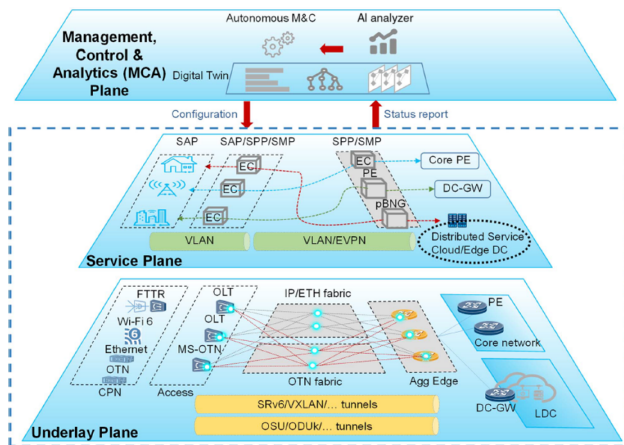


Fig. 1. ETSI F5G architecture [25].

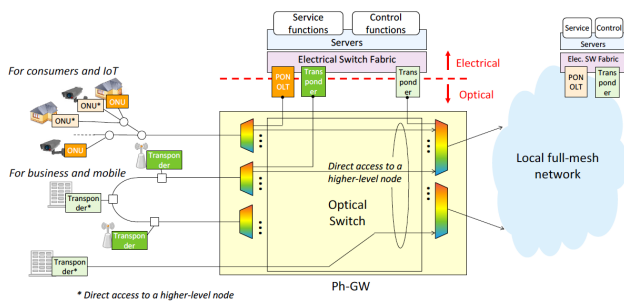


Fig. 2. Optical access network of the APN [28].

the Ph-GW and end user equipment (UE) is through a protocol, modulation, topology agnostic auxiliary management and control channel. ML can be harnessed to enhance the above capabilities of the Ph-GW. More specifically, by liaising with the control layer of the APN and through capacity and wavelength channel prediction, the Ph-GW can ensure that contention on the same wavelength channel is avoided and that end UE transceivers are proactively tuned to the designated wavelengths for transmission/reception.

Network 2030. The vision paper from a study initiated by the ITU-T Focus Group Technologies (FG NET-2030), Network 2030, stresses the significance of holographic-type communications (HTC) and multi-sense services as drivers of the development of next-generation networks over the next decade [29]. It is expected that such applications will evolve from current entertainment-focused to society-focused with superior engaged experience, e.g., remote holographic presence. As the complexity of successfully supporting HTC and multi-sense services is high, e.g., the need to deliver volumetric data from multiple viewpoints (tilt, angle, and position) that are also synchronized to image arrays, edge compute resources along with local service provisioning within the access network were highlighted as an agnostic service deployment solution. Crucially, FG-NET-2030 acknowledges the importance of ubiquitous fiber connectivity along with coordination of distributed intelligence to realize these applications, specifically providing fusion of digital and real worlds of machines, robots, and their virtual counterparts over the entire connected network fabric [29].

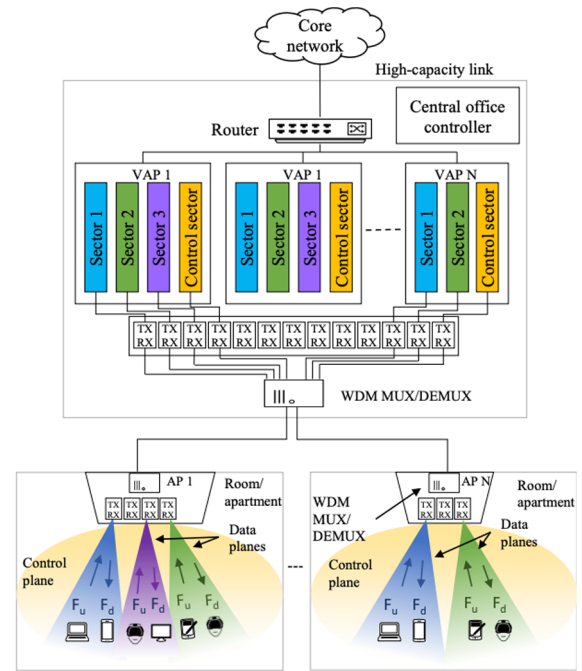


Fig. 3. Schematic diagram of the universal network [30].

Last Meter Fiber Wireless Integration. As with the complementary use of WiFi 6 in F5G, the exploration of alternative wireless network technologies for integration with the fixed fiber network is also underway. Among the candidate technologies investigated are WiFi 6 extended (WiFi 6E) and optical wireless communication (OWC) technologies. In comparison to WiFi 6, which uses the 2.4 GHz and 5 GHz bands, WiFi 6E also incorporates the use of the 6 GHz band, thus providing additional spectrum to alleviate congestion problems. OWC technology, on the other hand, encompasses many competing technologies including visible light communication, light fidelity, and near-infrared OWC for ultra-high-speed transmission. Drawing from the benefits of each of these technologies, a universal solution based on a high-capacity optical access distribution network for indoor applications was proposed in [30]. Figure 3 illustrates the universal architecture where the centralized controller at the central office (CO) uses ML to predict user demand. In turn, the controller instantiates virtual access points (VAPs) at the CO to remotely control the functionality of physical access points of each apartment/room and to dynamically allocate wavelength and bandwidth resources in response to changing user demand. Crucially, the architecture supports the interoperability between indoor wireless LAN and OWC technologies to simultaneously realize high-reliability, low-latency, and high-bandwidth connectivity.

Optical Transport for Next-Generation Mobile Networks. With fiber installations penetrating deeper towards the end UE and with speeds of 50 Gbps and beyond, PON technology is expected to facilitate the reach and provide the capacity required by 6G [8]. The business case to consider an optical rather than a radio access network (RAN) becomes even more compelling when spare fiber already available in installed optical access networks, e.g., in urban cities, is available for use [8]. Even for greenfield deployments, a fiber-to-the-home

(FTTH)-5G convergence study by FTTH Council Europe found that adding sufficient spare capacity to the initial build of an FTTH deployment to accommodate a five-year phased fiber-5G rollout incurs only a minimal additional investment (<1%) in return for future cost-savings in 5G rollout of up to 3.5 times [9]. According to [31], ML must play a critical role in all network layers and entities—from the management layer and physical layer to the radio base stations and mobile devices—to satisfy the unprecedented demand for high-bandwidth capacities (>1 Tbps), ultra-low latencies (~microsecond scale), ultra-high reliability (6 nines reliability), and massive connectivity over 3D coverage areas promised by 6G.

3. MACHINE LEARNING OVERVIEW

A. 21st Century Revival

ML has been a topic of research within statistics and computer science for the past 70 years, with the notion of AI being first introduced by Alan Turing in 1950. During the same decade, the first neural network (NN) for computers, known as the perceptron, was invented by Frank Rosenblatt. The term “machine learning” was introduced by Arthur Samuel in 1959 [32] defining the field of study in which computers can perform tasks without being explicitly programmed to do so. Since then, ML research has gone through periods of dormancy, known as “AI winters” in the early 1970s and late 1980s, and as many revivals—we are currently in the midst of a resurgence. The current intense interest in ML technologies and their applications is a confluence of the critical drivers below [33–35].

- **Data.** With the rise of the Internet, Internet-of-Things, social media, big data, and network equipment with advanced data acquisition and monitoring capabilities, datasets for training, testing, and validation are now becoming more accessible than ever. Coupled with the emergence of cloud and multi-access edge computing, the processing and storage of comprehensive datasets have also become more cost efficient rationalizing ML use.

- **Open-source software and hardware advances.** Over the past decade, powerful and open-source toolkits, libraries, and frameworks have led to the accelerated innovation of ML technologies and applications and faster adoption in a wide range of disciplines. In terms of hardware, the emergence of graphics processing unit (GPU) based computing and hybrid field programmable gate arrays is accelerating NN processing capabilities. Such capabilities not only accommodate large datasets, e.g., via dedicated video random access memories in GPUs but also realize multiple, distributed, and simultaneous computations, thus accelerating processing and analytical capabilities without incurring additional power.

- **Cloud and multi-access edge computing.** Previously, running ML capabilities required the use of expensive, dedicated, and exclusive supercomputers. ML has now become democratized and more cost efficient with the advent of cloud and multi-access edge computing [33]. This, in turn, is generating more interest from a wider segment of the general population, and the application of ML is evolving from solving

problems beyond simple data analytics into predictive and prescriptive analytics.

- **Investments.** Industry, governments, and research consortia are significantly investing people capability and research dollars into ML technologies and in their applications. According to the International Data Corporation [36], global spending on AI will grow from \$85.3 billion in 2021 to more than \$204 billion in 2025, representing a compound annual growth rate of 24.5% over the 2021–2025 forecast period. Compared to previous revivals, ML technologies and their applications are now being developed and tested in a more collaborative way among industry, government, and the research community, with the core goal to accelerate migration towards deployment.

B. ML Terminology and Algorithm Classification

Figure 4 illustrates the terminology and algorithm classification of ML. While the terms ML and AI are often used interchangeably, ML is by definition a subset of AI. ML algorithms enable computers/machines to learn from data and improve from experiences to accomplish a particular task without having to be programmed. In comparison, AI algorithms and techniques enable machines to simulate human behavior such as reasoning, sensing, reacting, and adapting over time when performing tasks. Within ML, its algorithms can be classified as follows [37,38].

- **Supervised learning.** Algorithms train with a labeled input dataset, i.e., “ground truth” to build a mathematical model that maps inputs (features) to the desired output (label). Supervised learning is mainly used to solve regression problems whereby an output of a future observation is predicted or a classification problem where the output is derived from one or more classes or groups. Regression outputs are generally continuous values, whereas classification outputs are discrete.

- **Unsupervised learning.** In unsupervised learning, algorithms train using unlabeled data with the aim to learn patterns or behaviors that are similar, e.g., in clustering, or spot exceptions in the dataset, e.g., in anomaly detection.

- **Semi-supervised learning.** A hybrid of the previous two categories, semi-supervised learning algorithms utilize unlabeled data to improve the accuracy of supervised learning models, trained using a small set of labeled data. Semi-supervised learning is commonly used when labeled datasets are limited, e.g., in network attack detection where it is infeasible to have a complete representative dataset with labels for all possible attack scenarios.

- **Reinforcement learning (RL).** This is a class of algorithms that do not require training data and instead rely on an agent who learns how to optimally behave through a “policy” of maximizing a reward. Specifically, the agent will interact with the environment, receive feedback after each action, and update its states accordingly, thereby self-improving over time.

- **Deep learning.** Cross-cutting the above categories of ML algorithms is deep learning, which uses artificial NNs (ANNs) that are inspired by the way the human brain processes. Specifically, ANNs adopt a layered learning architecture comprising neuron units much like the network of neurons

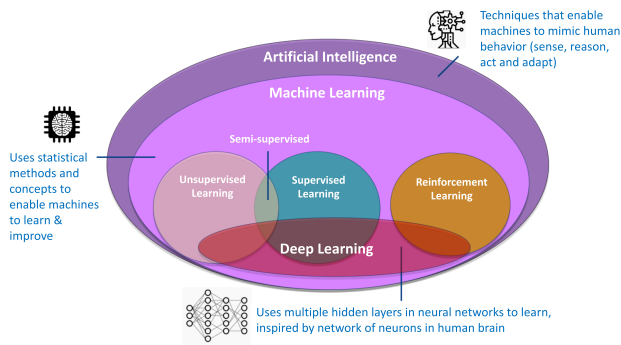


Fig. 4. ML terminology and algorithm classification [2].

in the human brain, contributing to its capability to learn and characterize highly complex patterns and nonlinear structures of data addressing a variety of problems including regression, pattern recognition, and classification. The term “deep” refers to the use of multiple hidden neuron layers in the architecture of the deep NN (DNN).

4. MACHINE LEARNING CHALLENGES

This section provides a summary of some open key challenges that are impeding the progress of ML models to production. A visualization of these challenges and potential ML approaches to address them are shown in Fig. 5.

Availability of representative data. In optical communication, obtaining sufficiently detailed datasets from deployed networks remains a significant challenge, as it is difficult to collect experimental data and practical data from network operators or corporations. Data may not be accessible due to a lack of monitoring infrastructure. As importantly, it could be cost prohibitive to retrieve and label the required volumes of data. There could also be a lack of expertise to label data. Figure 6 illustrates the relationship between the likelihood of ML adoption against the cost per labeled sample. As can be inferred, the motivation to use ML is high if a sufficiently detailed and effective dataset that is representative of the problem to be solved can be generated at a low cost [39].

Emerging approaches to overcome the lack of representative data include *generative adversarial networks (GANs)*, which were initially introduced to augment and increase the size of small labeled datasets for vision and image processing [40], and *active learning*, which is useful in situations where there exist large volumes of unlabeled data but labeling them is time

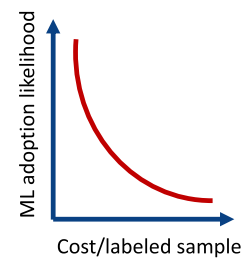


Fig. 6. ML adoption likelihood against the cost/labeled sample.

consuming and/or expensive [41]. Another approach is *online learning*, which involves offline training with a small training dataset and then continual learning when data becomes available during operation [42]. *Transfer learning* is another candidate to consider when there is a lack of representative data. Transfer learning allows a target ML model to reuse the parameters of a source ML model to perform a target task. In doing so, only a small training dataset is required to optimize the target ML model [43].

Adaptability to dynamic network environments. In ML research, it is common to assume that data are stationary and identically distributed. That is, the distributions of training datasets and future datasets in deployment are identical, and the function that is being learned is stationary for a sufficiently long time. However, in most real-world deployments, networks can become dynamic over time and/or can evolve to carry different types of traffic. In such cases, the underlying distribution of the training data is no longer representative of that in deployment. This phenomenon is known as *concept drift*, and when it happens, the learned function is no longer accurate and will result in a poor prediction/decision [44]. Dynamic networks have thus driven research towards adaptable ML models and *concept drift detection*. Online learning is an ML technique that is able to accommodate changes in data distribution, data types, features, etc., when in operation. This is because the model processes data samples in real time during operation [42]. Similarly, *federated learning* is another approach whereby it combines the local learning of many decentralized local ML models and shares this learning globally to improve the adaptability of each local ML model [45]. Additionally, the need to support real-time applications is further placing additional demands on ML models to re-train with high responsivity. In that respect, using computationally efficient ML techniques will incur less training time. Candidates include online learning, which processes only a small amount of samples at a time, federated learning, which distributes learning across the federation and thus reduces training time at each local node, and transfer learning, which uses a small target dataset to allow for rapid adaptation and performance improvement of a target task.

Explainability and trust. Despite widespread demonstrated performance improvements, ML models are considered “black boxes,” a term used to explain the little to no understanding of the reasons behind predictions/decisions made. To date, optical network and systems planning and operations are overseen by domain experts. The experts need to be confident of the decisions and recommendations made by ML models, especially when pertaining to networks that support services that require

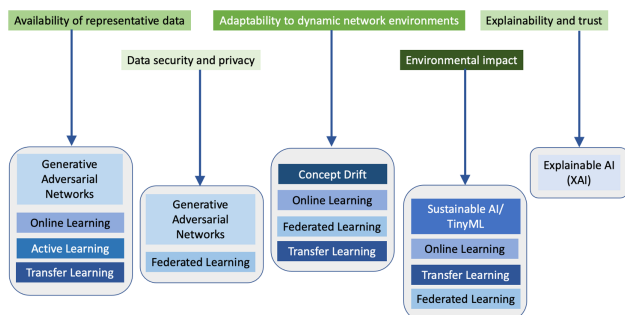


Fig. 5. Summary of ML challenges (green) and emerging approaches (blue).

very high levels of reliability. The push to progress ML models to production thus requires them to be explainable and their output interpretable. To this end, there is a growing field in *explainable AI (XAI)* where new techniques and principles are being developed to improve the explainability of ML models, so that their predictions and decisions can be better understood by humans [46]. Other associated matrices that are being investigated in XAI include trustworthiness (confidence in a model acting as intended), stability (whether the performance of the ML model remains stable to outliers), robustness (whether the ML model can tolerate perturbations that human beings have deliberately produced), and reproducibility (whether an ML model can reproduce the same results if the same dataset is used).

Environmental impact. Assessments of the carbon footprint of ML models have pointed to the relationship between their computational complexity and carbon emissions [47,48]. As an example, the report in [47] alarmingly highlights that common ML models used for natural language processing can cause significant amounts of CO₂ emissions. Pointedly, the report states that the CO₂ emissions from training one NN model with a GPU can be as high as the lifetime CO₂ emission of five cars. As a counterpoint to promote sustainable AI, a list of recommendations was presented in [48], which includes using computationally efficient ML models alongside energy-efficient hardware.

Data security and privacy. ML is not without its own security challenges, the foremost being the manipulation of training data to yield erroneous decisions/predictions. Malicious attacks may also range from data extraction whereby the integrity and privacy of the training data are undermined, through to transfer learning attacks whereby the source model is compromised. Currently, a vast body of work from diverse scientific communities paints a picture that many vulnerabilities of ML and the countermeasures used to defend against them are beginning to mature. GANs, initially proposed for data augmentation, are now finding use in security applications [49]. In addition, collaborative ML models are also being explored to de-risk single points of attacks. Seeing that a central data collection, learning, and inference location, e.g., at the CO, increases the vulnerability of successful attacks, collaborative models that distribute data collection, learning, and inference locations can alleviate this vulnerability. Federated learning is an example of a collaborative learning model that uses this distributed concept. Apart from removing a central point of vulnerability, federated learning also benefits from improved data privacy, as only local model parameters, rather than local datasets, are shared across the federation [45].

5. EMERGING ML APPROACHES

This section presents an overview of emerging ML approaches that are currently being explored to address the aforementioned challenges. Where applicable, examples relevant to optical access networks will be highlighted.

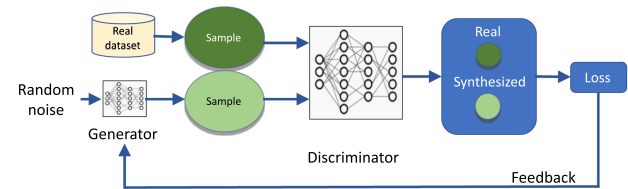


Fig. 7. Schematic of the GAN.

A. Generative Adversarial Network

GANs were initially proposed to synthesize data to augment and increase the size of small labeled datasets in computer vision and image processing applications [40]. As shown in Fig. 7, the framework of the GAN consists of two NN models: a generator that captures the distribution of real data to generate new data and a discriminator that decides whether a sample is real or generated [40]. During training, the two models engage in a game whereby the generator's aim is to generate data so real that it is indistinguishable to the discriminator, whereas the discriminator's aim is to accurately classify real data from generated data. At the end of the training phase, both models would be optimized such that the generated data and actual data are indistinguishable [40].

In its application to optical communication systems and networks, the authors of [50,51] proposed a network traffic data augmentation technique based on a GAN to adaptively augment the traffic dataset for various scenarios. The authors of [52] utilized a new GAN structure to learn inter-symbol interference. In [53], the authors employed GANs to model an optical intensity modulation/direct detection system. In [50,51], the authors used GANs to model the optical fiber channel with channel impairments including chromatic dispersion, self-phase modulation, attenuation, and erbium-doped fiber amplifier (EDFA) induced amplified spontaneous emission noise.

GANs are also finding use in security applications as previously discussed [49]. Though existing solutions have been specifically proposed for wireless networks, these can be potentially used to improve existing defense strategies in the optical domain. In the wireless domain, the authors of [54] proposed a modified bidirectional GAN architecture to evaluate an anomaly score for anomaly detection. In [55], GANs were proposed for spoofing through generating synthetic wireless signals that are indistinguishable from true signals. Conversely, in [56,57], GANs were used as a defense mechanism by countering adversarial attacks.

B. Active Learning

As with GANs, active learning is helpful when labeled data are difficult to obtain and/or when the size of the labeled dataset is constrained. It is also useful when labeling data is time consuming and/or expensive. In active learning, an ML model first trains on a labeled or "seed" dataset, and then it proactively queries an oracle to label unlabeled samples [42]. The oracle can be a human, a database, or software. With each new labeled sample, the model subsequently updates its activation function, and the process is repeated. Like semi-supervised learning, active learning models are trained using both labeled

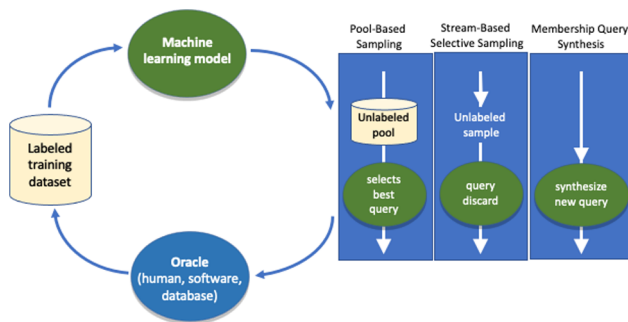


Fig. 8. Schematic of the active learning model.

and unlabeled data, but unlike semi-supervised learning, the unlabeled data in active learning are selectively, dynamically, and incrementally labeled. Overall, active learning draws upon selective query and iterative labeling to yield a high-accuracy model that uses fewer labeled samples and less time. Figure 8 shows the three categories of active learning, namely, stream-based or sequential selective sample, pool-based sampling, and membership selective synthesis. In stream-based selective sampling, the model is sequentially presented with an unlabeled data sample and immediately decides whether it wants to query the label or discard it. In pool-based sampling, the model selectively queries samples to be labeled from a pool of unlabeled data. In membership selective synthesis, the model can query any unlabeled samples including those it synthesizes during the training phase [42]. Which unlabeled sample to query depends on the strategy used. Example strategies include uncertainty sampling, which queries samples that the model is least certain on how to label, and variance reduction, which queries samples that minimize the future error by minimizing its variance [42].

To date, in optical communication systems and networks, the use of active learning has been focused on selecting and extending training datasets with the view to improving prediction accuracy for optical network management purposes. In [58], three pool-based active learning query strategies were studied to collect training data points of an EDFA model to improve channel gain prediction. In [59], the performance of active learning and domain adaptation, a subclass of transfer learning, was studied from the context of estimating the signal-to-noise ratio (SNR) using additional samples acquired from selected probe lightpaths in flexible grid networks. Both models achieved comparable prediction accuracy to supervised learning but benefit from a smaller dataset. In fact, active learning was shown to require approximately an order of a magnitude fewer additional samples than that required by domain adaptation.

C. Online Learning

Online learning involves training offline before deployment and continuously learning in real time while in the field by retrieving new samples to improve its current model, as shown in Fig. 9. Online learning is thus suitable when training data are limited during the training phase but are expected to become available over time in deployment [41]. Not only is it data and space efficient due to the processing of only small

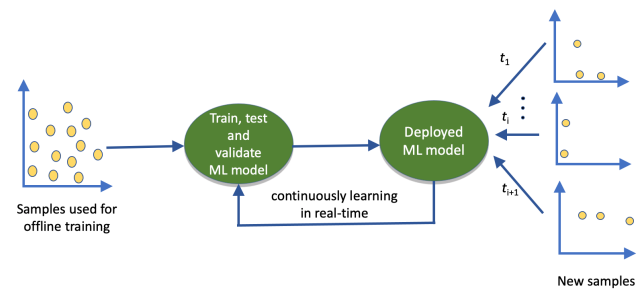


Fig. 9. Schematic of the online learning model.

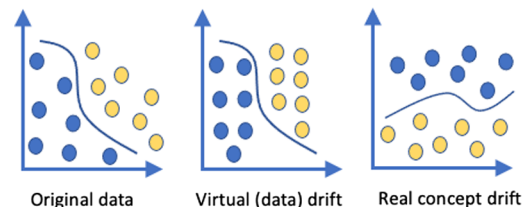


Fig. 10. Impact of virtual (data) and real concept drift on the decision boundary.

amounts of data samples at a time, but it is also adaptable and time efficient, using only the latest samples to learn. The downside to online learning is that in deployment, the model must be able to detect and adapt to concept drift and consequently be able to decide which learned knowledge to retain and which to replace with new information. Concept drift and its detection techniques are discussed next.

D. Virtual (Data) and Real Concept Drift Detection

Concept drift detection is important to the retraining process of ML models. There are two types of concept drifts, namely, data drift (also known as virtual concept drift) and real concept drift. In data drift, the distribution of the input data changes without affecting the decision boundary. Data drift is caused by data variation between the training and production phases, e.g., due to sample selection bias, thus undermining the performance of the ML model. In general, data drift detection techniques are typically based on monitoring input data distribution changes and are simpler than real concept drift detection techniques.

In comparison, real concept drift changes the underlying functional relationship between its inputs and outputs, as illustrated by the change in the decision boundary in Fig. 10. To detect real concept drift, an online error rate monitor in conjunction with a threshold value was proposed in [44]. To specifically target gradual drifts over time (as opposed to sudden drifts), a method using the distribution of distances between errors was proposed in [60]. In [61], a method using a combination of past and current models was proposed. In general, the methods mentioned above require dedicated storage and processing capabilities. To overcome this limitation, the authors of [62,63] utilized federated learning to distribute and scale computation resources. Model parameter updates sent by federated nodes during each iteration of the federated learning process are compared and used to detect real concept drift.

E. Explainable Artificial Intelligence

The urgent push to progress ML models to production is fueling intense interest in XAI. XAI techniques can be categorized into (a) model-agnostic approaches and (b) model-specific intrinsic approaches [46,64,65]. The former can be applied to any ML model in a *post hoc* manner, and examples include local interpretable model-agnostic explanations (LIME) [66], Shapley values, and Shapley additive explanation (SHAP) [67], which can be used to explain any ML model. In comparison, the gradient class activation mapping (GRAD-CAM) and deep learning important features (DeepLIFT) techniques are specific to DNN models [68,69]. Due to the flexibility and ease of use, research has been focused on developing flexible model-agnostic approaches rather than model-specific methods [69]. In XAI, the outputs can be visual, textual, or mathematical/numerical and can be classified as explanations and/or interpretations [65]. Explanations provide meaningful information that is comprehensible to humans. By contrast, interpretations reveal the extent, type, or which part of the data that has influenced the decision(s). Interpretations may or may not be understood by humans.

The use of XAI to explain black-box ML models in optical communications and networks has been limited to date, but preliminary investigations are gaining momentum. In [70], the authors employed the model agnostic XAI technique of SHAP to identify features that are most relevant to lightpath quality-of-transmission (QoT) estimation. The goal is to determine whether there exist relevant features that are common across two different synthetic datasets and across different classification models, and to investigate the dependency of lightpath QoT estimation on these features. Further, the authors of [71] used XAI in a network failure localization problem whereby SHAP was employed to identify correlations between the input data and model decisions and compare explanations in two network scenarios of differing optical SNR monitors along a lightpath. Using XAI, visualization results show that additional monitors would improve model explainability and trust and thereby the adoption of ML tools for fault management more generally.

F. Sustainable AI

Using computationally efficient ML models along with energy-efficient hardware are two measures recommended to reduce the carbon footprint of AI [48]. As already discussed, online learning, transfer learning, and federated learning are candidate ML models that are storage and computation friendly. With regards to energy-efficient hardware, TinyML is being explored to reduce CO₂ emissions. TinyML research involves investigating ML models that can be potentially computed on low-powered and small footprint devices such as microcontrollers, which consume six to eight orders of magnitude less power than GPUs. Moreover, placing these microcontrollers together with edge clouds will also realize low-latency and low-bandwidth model inference since data need not be transmitted to the cloud. In an enterprise network deployment, running TinyML at the edge also better preserves data privacy and integrity. Undoubtedly, the sustainable AI movement is gaining traction with the holistic view of taking into account

the ecological, environmental, and social impacts of the entire lifecycle of AI products and their implementation [72].

G. Federated Learning

Proposed in 2019, federated learning is a collaborative learning mechanism that enables local ML models executed at edge devices/servers to periodically share their model parameter(s) without compromising the privacy of their data [45,73,74]. As illustrated in Fig. 11, each local ML model would periodically share its local parameter(s) with a centralized cloud server that, in turn, would update a set of global model parameters. The global parameters would then be used to update each local node. In the event of concept drift, periodic global updates are expected to improve the adaptability of each local ML model.

Within access network research, a bandwidth slicing mechanism was reported in [75] to support the periodic upload of local model parameters from edge devices serviced by a PON and the download of global model parameters from the central server located at the CO. The slicing mechanism reserves and allocates dedicated upstream bandwidth for the transmission of local model parameters. Due to limitations in the amount of bandwidth that can be reserved, the number of supported edge devices is constrained. To overcome this limitation, the authors of [76] proposed a two-step aggregation method. Instead of transmitting directly to the CO, local model parameters of edge devices are first aggregated at the optical network units (ONUs). Thus, upstream bandwidth reservation is constant irrespective of the number of supported edge devices. Further, to simultaneously deliver federated learning traffic and traditional broadband applications over the same 50G-EPON infrastructure, the authors of [77] proposed a dynamic wavelength and bandwidth allocation scheme along with two prioritization policies to reduce the upstream latency of federated learning traffic and delay-critical applications over the same network.

Aside from maintaining local data privacy, federated learning is also computation and time efficient. By sharing local model parameters through periodic global model parameter

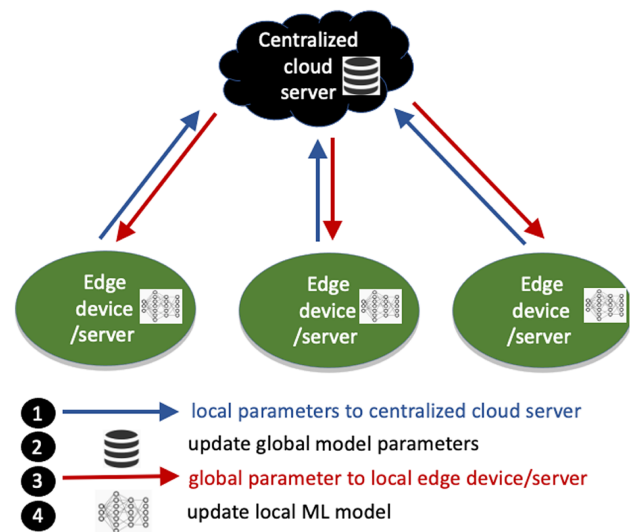


Fig. 11. Federated learning mechanism between the centralized cloud server and edge devices/servers.

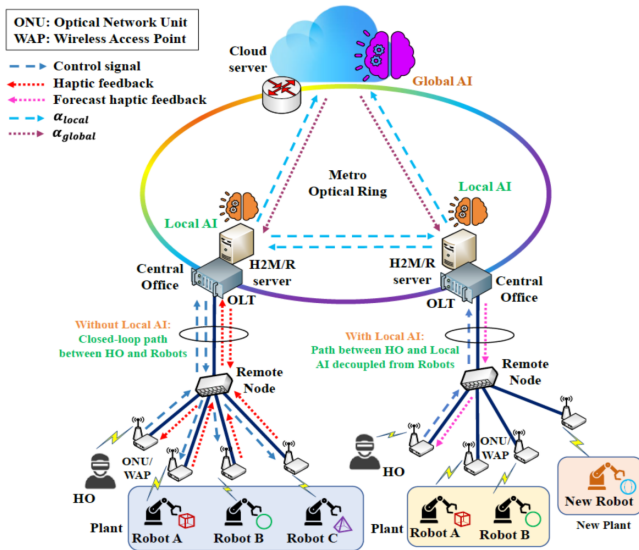


Fig. 12. Local and global AI servers supporting H2M/R collaboration over converged access networks [78].

updates throughout the federation, computation and time savings can be achieved. As an example, in [78], we proposed a coordinated learning framework based on federated learning such that local servers can dynamically support the rapid onboarding of newly introduced machines/robots. As shown in Fig. 12, local servers, each supporting a human operator and one or more machines/robots, collaboratively learn from each other and also from a global AI server. Without federated learning, and as the number of machines/robots connected to the local AI server is increased, more iterations of exploration/exploitation in RL, and hence learning time, are required to find the optimum local parameter [79]. This is in contrast to when federated learning is implemented, whereby the local model is updated with the global parameter. Results in [78] highlight that a time savings of up to 75% can be achieved to find an optimum local parameter in a network where four machines/robots collaboratively learn from each other and from the global AI server.

Similarly, in [80], we proposed a federated learning framework that harnesses shared learning between multiple converged access networks to expedite bandwidth allocation decisions in support of latency-sensitive H2M/R communications. Specifically, the framework combines RL to optimize local bandwidth allocation decisions and share this local learning with a federation of PONs to expedite bandwidth allocation decisions. Results show that using federated learning, a significant convergence time savings of up to 80% can be achieved in a federation of five PONs.

H. Transfer Learning

In transfer learning, knowledge is transferred from the source domain to the target domain. Specifically, the parameters of an existing ML model (termed the source model) trained on source data to perform a source task are re-used to train another ML model (termed the target model) on a different target task, as illustrated in Fig. 13 [43,81]. As such, the computational effort and amount of training data required in the

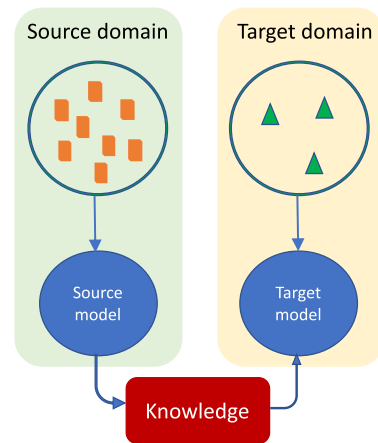


Fig. 13. Schematic of transfer learning process between the source and target domains.

target domain are reduced, and only a smaller target dataset is required to refine the target ML model. Hence, transfer learning can be considered as an optimization tool to allow for rapid adaptation and performance improvement of the target task.

Unlike federated learning, transfer learning was proposed almost two decades ago with its early use extensively in data mining [43]. It was only in recent years that interest in the use of transfer learning in optical communication systems and networks was piqued. This has been predominantly due to the lack of representative data and the need to adapt to increasingly dynamic networks. As an example, the lack of failure data (due to the low probability of network failures) in deployed optical networks motivated the exploration of transfer learning in failure detection and failure cause identification [82,83]. In estimating and predicting QoT, transfer learning between multi-domain elastic optical networks was explored in [84] to optimize DNN model parameters. In [85], transfer learning was investigated to reduce the uncertainty of the generalized SNR to improve the precision of the QoT estimation of lightpaths. In terms of addressing network dynamicity, transfer learning was investigated in the context of elastic optical networks to expedite routing, modulation, and spectrum allocation decisions and thus non-blocking performance in [86]. In terms of short-reach optical links, transfer learning was investigated in [87] to expedite the training time of feedforward NN and recurrent NN-based equalizers of 50 Gbps pulse amplitude modulation 4-level direct detection 20 km optical links. In the next section, we will showcase the time-saving benefit of transfer learning through a novel self-adaptive DBA scheme that accommodates traffic pattern and network load changes in next-generation MFH networks.

6. SELF-ADAPTIVE MOBILE FRONTHAUL BANDWIDTH ALLOCATION

Next-generation mobile networks are evolving towards 6G and are expected to use higher carrier frequencies, thus necessitating shorter-ranged, smaller-sized, and highly dense cells [8,9]. As a candidate for MFH, PON technology will provide the cost benefit, reach, and capacity required. Moreover, with

a P2MP physical topology, PON lends itself naturally to the statistical multiplexing of the remote radio units (RUs) of the centralized RAN architecture. In this architecture, central/distribution units (CUs/DUs) are physically co-located with the optical line terminal (OLT) of the PON. In turn, the CU/DU-OLT is directly connected via a passive fiber plant to ONUs that are physically co-located with RUs [88]. With the PON serving as the MFH network, uplink bandwidth is shared and hence uplink latency performance is highly dependent on how bandwidth is allocated to each ONU-RU for uplink transmission. Further, with the expectation to support a range of diverse and immersive services, the MFH network must be able to rapidly adapt to a dynamic network environment that supports immersive 6G experiences. Here, we propose an FSA ML-enhanced DBA (FSA-DBA) that combines RL and transfer learning to achieve expedited self-adaptive and optimized bandwidth decisions.

A. Fast Self-Adaptive DBA for Mobile Fronthaul

In an MFH network, a UE will request for transmission by sending a scheduling request to the CU/DU. Subsequently, the CU/DU grants radio resources by transmitting a scheduling grant back to the UE. Innovative bandwidth allocation schemes to satisfy the uplink MFH bandwidth and latency requirements have been previously proposed. These include cooperative DBA schemes that use fronthaul traffic information through a cooperative transport interface message [89] to improve overall latency and bandwidth utilization, e.g., [90,91]; priority scheduling DBA based on traffic load [92]; bandwidth guaranteed DBA based on network slicing of mobile and other services [93]; and self-adjusting DBA that dynamically adjusts fronthaul allocation intervals based on reports [94]. The amount of allocated bandwidth, termed T_{grant} , to each ONU-RU is estimated based on packet arrival traffic characteristics at the ONU-RU. Specifically, in [93,95,96], T_{grant} is estimated based on the Common Public Radio Interface constant bit rate traffic with a fixed packet size. Conversely, in [91,97,98], T_{grant} is estimated using packet arrival distributions at ONU-RUs and inter-packet gaps in MFH transmission. Unlike the discussed MFH-DBAs, our proposed FSA-DBA scheme does not require prior knowledge of MFH traffic to achieve T_{grant} decisions.

Self-adaptive decision. FSA-DBA leverages RL to estimate self-adaptive T_{grant} decisions that are best suited for the underlying traffic. To reduce latency, the reward R is equivalent to the negative latency measured at each ONU-RU as a result of a T_{grant} allocation. Subsequently, for the CU/DU-OLT to self-adaptively adjust T_{grant} for that ONU-RU, a decision-value function, denoted Q_{target} , associates T_{grant} with R . So, considering $T_{\text{grant}}(t)$ and $R(t)$ to represent the bandwidth decision and reward, respectively, in the t th iteration, Q_{target} is updated as follows:

$$Q_{\text{target}}(T_{\text{grant}}(t)) \leftarrow (1 - \alpha) Q_{\text{target}}(T_{\text{grant}}(t)) + \alpha(R(t) + \gamma \max Q_{\text{target}}). \quad (1)$$

Parameters α and γ ($0 \leq \alpha, \gamma \leq 1$) control the learning rate and the weight on future rewards, respectively. With each

ONU-RU reporting R to the CU/DU-OLT every K grant cycles, i.e., one iteration, this iterative learning and decision adjustment process thus realizes a self-adaptive and optimal T_{grant} decision for each ONU-RU.

Fast decision exploration. A drawback of RL is that its decision exploration can incur a time cost in converging to the optimal T_{grant} decision. To this end, FSA-DBA leverages transfer learning to reuse existing decision-value knowledge to expedite learning. Defining $\mathbf{Q} = \{Q_{\text{source}, n}\}$ to be a set of source decision-value set pre-acquired through either simulations or past MFH operations, FSA-DBA identifies the source knowledge, indexed by n^* , that is most related to the learning of Q_{target} as follows:

$$n^* \leftarrow \operatorname{argmax}_n |Q_{\text{source}, n} - Q_{\text{target}}|. \quad (2)$$

The distance $|Q_{\text{source}, n} - Q_{\text{target}}|$ in Eq. (2) implies the similarity between the source domain where $Q_{\text{source}, n}$ is obtained and the current target domain [17]. Once Q_{source, n^*} is known, this is used to guide the decision exploration in a neighborhood range $D = [T_{\text{grant}, n^*} - \delta, T_{\text{grant}, n^*} + \delta]$ to $T_{\text{grant}, n^*} = \operatorname{argmax} Q_{\text{source}, n^*}$. For fast convergence, we adopt the greedy policy to adjust T_{grant} , whereby

$$T_{\text{grant}} \leftarrow \operatorname{argmax}_D Q_{\text{target}}. \quad (3)$$

Overall, during network operation, an ONU-RU will report $R(t)$ every K cycles to the CU/DU-OLT, which will then update Q_{target} following Eq. (1). Subsequently, T_{grant} is adjusted by referring to the source decision-value set $\mathbf{Q}$ following Eqs. (2) and (3). With Q_{source, n^*} to guide the decision exploration, the learning time to reach an optimum decision, T_{grant} , can be reduced.

B. Performance Evaluation

Packet-driven simulations were performed to validate the latency and learning rate of FSA-DBA. A 10 km 10G-EPON (IEEE Std 802.3 av [99]) with 16 ONU-RUs that supports an MFH with 500 Mbps radio access capacity was considered in our simulations. Further, packet arrivals with varying packet sizes from 64 to 1518 bytes were either exponentially (EXP) [92,93], generalized pareto (GP) [100], and gamma (GA) [100] distributed. The normalized load of an ONU-RU was chosen to be from 0.1 to 1.0. This is equivalent to an aggregated MFH load of 0.08 (16 ONUs $\times$ 500 Mbps $\times$ 0.1/10 Gbps) to 0.8 (16 ONUs $\times$ 500 Mbps $\times$ 1/10 Gbps). The uplink latency performances of FSA-DBA, the cooperative DBA scheme with arrival estimation [97], and the baseline DBA scheme [95] were evaluated. The baseline DBA scheme refers to the conventional DBA scheme that relies on the communication of report and grant messages between the CO and ONUs to request and grant bandwidths, whereas the estimation scheme refers to the cooperative DBA scheme with arrival estimation every 50 μ s. The results are shown in Fig. 14. Unsurprisingly, the baseline scheme is shown to incur the highest latency since each packet needs to be first reported and then granted bandwidth. At minimum, the packets have to wait for approximately 100 μ s, corresponding to the round trip between ONU-RU and CU/DU-OLT. As the

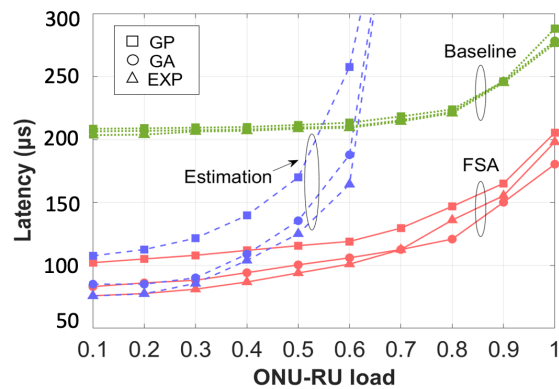


Fig. 14. Latency performance of FSA-DBA with different traffic patterns in simulation.

report-grant process is omitted from the cooperative DBA scheme, its uplink latency is lower than the baseline DBA scheme's. However, since the decisions for T_{grant} are based on the arrival estimation of fixed-length cycles (50 μs) [97], its results in Fig. 14 clearly reflect its lack of flexibility towards different traffic patterns and loads. Specifically with traffic loads above 0.6, longer grant cycles are needed to cater to the bandwidth demand. By contrast, the latency performance of FSA-DBA shows that it is able to self-adaptively adjust bandwidth decisions as well as grant cycles. Results highlight that FSA-DBA achieves the lowest uplink latency due to expedited self-adaptive and optimized bandwidth decisions. An uplink latency of approximately 200 μs is achieved under all three traffic patterns and network load scenarios, thus satisfying uplink MFH latency requirements [96,101].

The learning performance of FSA-DBA was then evaluated in a target network with empirical traffic traces previously collected from our immersive H2M application experiments [102]. The simulated Q functions of the aforementioned settings, i.e., the EXP, GP, and GA patterns, and the ONU-RU load 0.1 to 1.0 were used as source Q to assist in the learning of Q_{target} for empirical traffic. We further chose an iteration to be $K = 10$ numbers of grant cycles for ONU-RUs to report their rewards to the CU/DU-OLT and to adjust T_{grant} based on the updated Q_{target} . The learning rate and latency performance of the coordinated DBA scheme with arrival estimation, baseline DBA scheme, and FSA-DBA with and without transfer learning, i.e., using only RL, were compared, and the results are shown in Fig. 15. The plots show that FSA-DBA with and without transfer learning can achieve low latency. As importantly, with transfer learning, the FSA-DBA scheme requires only a few tens of iterations to explore an optimal decision compared to a few hundreds of iterations when not relying on transfer learning.

7. SUMMARY

We have witnessed the growth of optical access networks in the past 30 years, with continuing efforts in driving its capacity beyond 100 Gbps underway. We motivated its continued advancement and the rationale for adopting ML through a brief review of major consortia projects and industry trends in

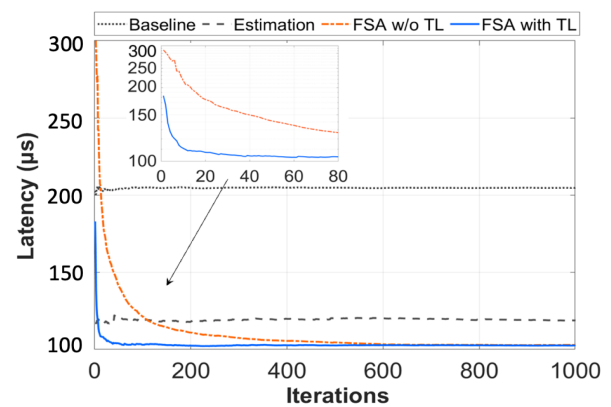


Fig. 15. FSA-DBA learning rate in empirical traffic with/without transfer learning (TL).

this tutorial. We also highlighted the enabling factors contributing to the current revival of ML research. As importantly, we outlined open key challenges facing the deployment of ML models in practice, especially focusing on those that are amplified by the traffic dynamicity and heterogeneity, data scarcity, and computation constraints of optical access networks. To counter these challenges, we highlighted emerging ML approaches that are currently being explored to minimize and/or to alleviate the impact of the outlined challenges. We then highlighted the benefits of one of these emerging approaches, namely, transfer learning, in an illustrative future use case of deploying PON to support immersive H2M traffic over the fronthaul of next-generation mobile networks. Specifically, we proposed a DBA scheme, namely, the FSA-DBA, that rapidly self-adapts bandwidth allocation decisions to achieve low-latency uplink transmissions in the MFH network. Without relying on any prior knowledge of traffic patterns, FSA-DBA explores an allocation decision best for the underlying MFH network. Results show uplink latencies of approximately 200 μs can be achieved with self-adaptive decisions for different traffic patterns and load scenarios. With PON as a potential fronthaul solution, FSA-DBA not only satisfies the uplink bandwidth and latency requirements of 6G immersive experiences, but also rapidly self-adapts to changing traffic conditions.

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